Dr. Shubham Dixit

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RESEARCH INTERESTS

My research focuses on developing process-informed statistical models for hydroclimatic extremes. My PhD examined how uncertainties arising from modelling choices, particularly threshold selection and climatic-covariate selection, can be reduced in nonstationary extreme-value analysis. This work led to a decomposed framework in which rainfall occurrence and severity are represented separately, allowing their distinct climatic controls and contributions to design return levels to be examined. My future research aims to develop physically informed nonlinear relationships that are more robust under changing climatic conditions and to extend this decomposition philosophy from rainfall to flood-generating processes.

PhD Thesis: Climate-Informed Return Level Estimation of Extreme Precipitation: A Diagnostic Non-Stationary Framework with Frequency-Severity Decomposition

PUBLICATIONS

Research arising from PhD thesis *(05 Published/Accepted; 02 under review/revision)*

1. Published/Accepted:

- **Dixit, S.,** & Pandey, K. K. (2026). Occurrence-rate nonstationarity alters design rainfall return levels: A decomposed Poisson-GPD analysis across contrasting hydroclimatic regimes regimes. *Journal of Hydrology*, 136627. https://doi.org/10.1016/J.JHYDROL.2026.136627
- **Dixit, S.,** & Pandey, K. K. A Diagnostic-Based Framework for Extreme Value Theory-Consistent Automatic Threshold Selection in Peaks-over-Threshold Analysis. *ASCE Journal of Hydrologic Engineering* (***Accepted and forthcoming***)
- **Dixit, S.,** & Pandey, K. K. (2026). "Refining Extreme Event Selection in Hydrological Analysis Using the Comparative Distribution Fitting Method.", *ASCE Journal of Hydrologic Engineering* 31.5 https://doi.org/10.1061/JHYEFF.HEENG-6733
- **Dixit, S.**, & Pandey, K. K. (2026), "Evolving Indo-Pacific Teleconnections and Regional Feedbacks in Indian Rainfall Extremes: A Principal Component-Wavelet Framework.", *Atmospheric Research*, *336*, 108900. https://doi.org/10.1016/j.atmosres.2026.108900
- **Dixit, S.**, & Pandey, K. K. (2024). "Spatiotemporal variability identification and analysis for non-stationary climatic trends for a tropical river basin of India.", *Journal of Environmental Management*, 365. https://doi.org/10.1016/j.jenvman.2024.121692.

2. Under Review:

- ***Preprint:*** **Dixit, S.,** & Pandey, K. K. "An Extreme-Value Framework for Separating Rainfall Occurrence and Severity: Regime-Dependent Insights for Risk Assessment." *[Research Square preprint]* https://doi.org/10.21203/rs.3.rs-10177938/v1. (*Under review: Water Resources Management*)
- Pandey, K. K., **Dixit, S.**, Verma, A., Prakash, H. A Review of Nonstationary Extreme Rainfall Frequency Analysis for Climate-Informed Estimation of Design levels. *ASCE Journal of Hydrologic Engineering* (Under Revision)

Other Publications:

- **Dixit, S.**, Roy, S., & Pandey, K. K. (2026). "Land-use transitions and hydrological futures in a monsoon-fed basin: insights from the Krishna River Basin, India.", *Journal of Water and Climate Change* 2026; jwc2026063. doi: https://doi.org/10.2166/wcc.2026.063
- **Dixit, S.**, Pandey, K. K., & Shukla, D. (2024). "Decadal hydrological impact assessment of evolving land use and land cover in an Indian river basin: a multi-model approach.", *Journal of Water and Climate Change*, jwc2024120. https://doi.org/10.2166/wcc.2024.120

- Vuppati, M.P., **Dixit, S.**, Deogade, R.B., Chandra, P. (2023). "River and Inlets Cross-Sections Extraction Using Stereo Pair Images for Hydraulic Modeling.", In: Timbadiya, P.V., Patel, P.L., Singh, V.P., Sharma, P.J. (eds) Hydrology and Hydrologic Modelling. HYDRO 2021. Lecture Notes in Civil Engineering, vol 312. Springer, Singapore. https://doi.org/10.1007/978-981-19-9147-9_24. **(Book Chapter)**
- **Dixit, S.**, G.K. Patil (2019). "Sediment Yield Analysis of Gangapur Dam using SWAT and GIS.", *Journal of Geotechnical Engineering (STM Journals)*, 6(2): 1–9p. https://doi.org/10.3759/joge.v6i2.3277.

EDUCATION

PhD in Hydraulics and Water Resources Engineering **Dec 2021 - Sept 2026**
Indian Institute of Technology (BHU), Varanasi, India

M.E. in Water Resources Engineering **2017 - 2019**
Government College of Engineering, Aurangabad, India

B.Tech. in Civil Engineering **2012 - 2016**
Noida Institute of Engineering and Technology, Greater Noida, India

AWARDS & ACHIEVEMENTS

- Recipient of the **Prime Minister's Research Fellowship (PMRF)** to pursue Ph.D. research at Indian Institute of Technology (BHU), Varanasi, India.
- Awarded with the **Student Travel Grant by Anusandhan National Research Foundation (ANRF)** to attend American Geophysical Union (AGU) Annual Meeting 2025 at New Orleans, Louisiana (USA).

PATENTS (INDIAN)

- **Published:** System and Method for Automatic Threshold Determination in Extreme-Value Statistical Analysis (*Application No. 202611001007*) ***(Inventors:** Dr. K. K. Pandey and **S. Dixit**)*
- **Filed:** Stability-based System and Method for Frequency–Severity Decomposed Covariate Prioritization in Non-stationary Extreme Value Analysis (*Application No. 202611064184)* ***(Inventors:** Dr. K. K. Pandey and **S. Dixit**)*

WORK EXPERIENCE

Central Water and Power Research Station, Pune, India **Oct 2019 - Jan 2022**
Junior Research Fellow

- Developed Hydrodynamic and Water Quality Models using DHI MIKE-11 for 127 Km stretch of Mutha River flowing through Pune City and Suburbs.
- Used CartoSat-1 stereo pair images for developing high resolution DEM and using the latter for extracting bathymetry data for river and its inlets.
- Collected in-situ water and wastewater samples and tested in laboratory for water quality parameters.

CONFERENCES and SYMPOSIUMS

- European Geosciences Union ***(EGU)***, General Assembly 2026, Vienna, Austria, May 2026 *(In-person poster)*
- American Geophysical Union ***(AGU)***, Fall Meeting 2025, New Orleans, USA, Dec 2025 *(In-person poster)*
- European Geosciences Union ***(EGU)***, General Assembly 2025, Vienna, Austria, April 2025 *(In-person poster)*
- Prime Minister's Research Fellowship ***(PMRF)*** Symposium, IIT-Hyderabad, March 2025 *(In-person poster)*
- ***HYDRO***-2024, CWPRS, Pune, December 2024 *(In-person oral)*
- American Geophysical Union ***(AGU)***, Fall Meeting 2024, Washington DC, USA, Dec 2024 *(In-person poster)*
- Prime Minister's Research Fellowship ***(PMRF)*** Symposium, IIT-Indore, March 2024 *(In-person poster)*
- HYDRO-2021 International Conference, SVNIT Surat, December 2021 *(Online)*

SKILLS

Technical skills:	GIS, SWAT, HEC-RAS, HEC-HMS, MIKE 11
Programming Languages:	Python, R, MATLAB, Latex
Languages:	English, Hindi

PEER REVIEW ACTIVITIES

Served as a reviewer for *ASCE Journal of Hydrologic Engineering, Journal of Environmental Management*, *International Journal of Climatology, Scientific Reports*, *Earth Systems and Environment* and *Physics and Chemistry of the Earth*.

WORKSHOPS DELIVERED AS INSTRUCTOR

As a part of the *deliverables* under *Prime Minister's Research Fellowship,* following hands-on training workshops were independently designed and delivered *(130+ hours)*:

- 30 hours of hands-on training on "**Introduction to QGIS and QSWAT with its application in Agriculture and Hydrology"** in department of farm engineering, Institute of Agricultural Sciences, Banaras Hindu University (BHU) from 25th February 2023 to 23rd April 2023.
- 30 hours of hands-on training workshop on "**Introduction to Python Programming and its application in Hydrology and Agriculture**" in the Department of farm engineering, Institute of Agricultural Sciences, Banaras Hindu University from 06th January 2024 to 09th March 2024.
- 23 hours of hands-on training workshop on "**Introduction to principles of GIS and Remote Sensing**" in the Department of farm engineering, Institute of Agricultural Sciences, Banaras Hindu University from 16th March 2024 to 30th March 2024
- 20 hours of hands-on training workshop on **"Introduction on Water Resources Engineering Tools and Software (QGIS, QSWAT and HEC-RAS)"** at Government College of Engineering, Aurangabad (MH) from 10th August 2024 to 19th October 2024.
- 20 hours of hands-on training workshop on **"Introduction on Water Resources Engineering Tools and Software (QGIS, QSWAT and HEC-RAS)"** at Rajkiya Engineering College, Churk, Sonbhadra (UP) from 02nd November 2024 to 29th December 2024.
- 10 hours of hands-on training workshop on **"Introduction to Python Programming and its application in Water Resources Engineering"** at Rajkiya Engineering College, Churk, Sonbhadra (UP) from 11th January 2025 to 09th February 2025.

TRAININGS and CERTIFICATIONS

- One-week high-end workshop on "**Challenges in Water-Related Disasters Risk Reduction: Vulnerability, Adaptation and Resilience Techniques**" (WRDRR) under the 'Karyashala' scheme of Accelerated Vigyan program of the Science and Engineering Research Board (SERB), Department of Science and Technology (DST), Government of India. held at **NIH-Roorkee** during January 16-21, 2023.
- One-week high-end workshop on "**STAtistical Analysis and Modelling of Climate Extremes**" (STACE) under the 'Karyashala' scheme of Accelerated Vigyan program of the Science and Engineering Research Board (SERB), Department of Science and Technology (DST), Government of india. held at **IIT-Indore** during December 12-17, 2022.
- Five days training course on **"Sediment Yield and Reservoir Sedimentation"**, National Institute of Hydrology (NIH), Roorkee under National Hydrology Project. 2022
- Ten days purpose oriented training program on **"Introduction to Python Programming and its Applications in Water Resources Sector"**, National Water Academy, Pune. 2021
- One-week short term training program on **"Quantum GIS and QSWAT applications in Water Resources Engineering"**, Government College of Engineering, Aurangabad 2018.